

# Mathematics for Computing

## Official Statistical Machinery & Module 3: Descriptive Statistics

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# Module Outline

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## Overview & Mission

MoSPI is the apex executive body responsible for the development, coverage, quality, and dissemination of official statistics in India, as well as monitoring major national development projects.

### 1. Statistics Wing (NSO)

- **Central Statistical Office (CSO):**  
Coordinates statistical activities, National Accounts (GDP), Industrial Statistics.
- **National Sample Survey Office (NSSO):**  
Conducts large-scale socio-economic sample surveys nationwide.

### 2. Programme Implementation Wing

- **Infrastructure Monitoring:** Tracks Central Sector Projects (> Rs.150 Crore).
- **Twenty Point Programme (TPP):**  
Monitors key poverty alleviation schemes.
- **MPLADS:** Oversees the MP Local Area Development Scheme.

## Overview & Organizational Structure

DES Kerala is the state's nodal agency responsible for the systematic collection, compilation, analysis, and dissemination of official statistics for evidence-based governance and state planning.

## Administrative Hierarchy

- **Headquarters (Directorate):** Vikas Bhavan, Thiruvananthapuram.
- **District Level:** 14 District Offices, each headed by a Deputy Director.
- **Sub-District / Field Level:** 61 Taluk Statistical Offices managing data collection across Municipalities and Panchayaths.

# Overview of Descriptive Statistics

## What is Descriptive Statistics?

Descriptive statistics involves summarizing, organizing, and presenting data in a meaningful way using numerical measures, tables, and graphical displays.

### Descriptive Statistics

- Focuses on summarizing observed sample data.
- Uses measures of central tendency and dispersion.

### Inferential Statistics

- Makes predictions about a population based on a sample.
- Uses probability models.

## Example: Descriptive vs Inferential

**Descriptive:** The average score of 50 students in this class is 78%.

**Inferential:** Based on this class's sample average, we estimate with 95% confidence that the average score of all 500 students in the university is between 75% and 81%.

# Histograms

## Definition (Histogram)

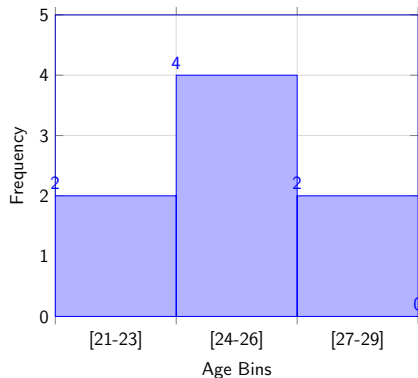
A **histogram** is an approximate representation of the distribution of numerical data. It partitions the spread of the data into contiguous intervals called **bins**.

## Example: Constructing a Histogram

**Data (Ages):** 22, 23, 24, 25, 25, 26, 28, 29

**Bins:** [21 – 23], [24 – 26], [27 – 29]

- Bin [21 – 23]: Contains 22, 23 (**Freq** = 2)
- Bin [24 – 26]: Contains 24, 25, 25, 26 (**Freq** = 4)
- Bin [27 – 29]: Contains 28, 29 (**Freq** = 2)



# Sample Mean, Variance and Standard Deviation

## Definition (Sample Mean and Variance)

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad \text{and} \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

## Example: Calculating Mean and Variance

**Data:** 2, 4, 6 (Sample size  $n = 3$ )

**Mean ( $\bar{x}$ ):**

$$\bar{x} = \frac{2 + 4 + 6}{3} = \frac{12}{3} = 4$$

**Variance ( $s^2$ ):**

$$s^2 = \frac{(2-4)^2 + (4-4)^2 + (6-4)^2}{3-1} = \frac{(-2)^2 + 0^2 + 2^2}{2} = \frac{4 + 0 + 4}{2} = 4$$

**Standard Deviation ( $s$ ):**  $\sqrt{4} = 2$ .

# Order Statistics

## Definition (Order Statistics)

When sample observations  $x_1, x_2, \dots, x_n$  are arranged in ascending order, the resulting ordered values  $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$  are known as order statistics.

## Example: Finding Order Statistics

**Raw Data:** 12, 7, 9, 3, 15

**Ordered Data:** 3, 7, 9, 12, 15 ( $n = 5$ )

- **1st Order Statistic (Minimum):**  $x_{(1)} = 3$
- **5th Order Statistic (Maximum):**  $x_{(5)} = 15$
- **Median (Middle value,  $x_{(3)}$ ):** 9
- **Range:**  $x_{(5)} - x_{(1)} = 15 - 3 = 12$

# Sample Covariance and Covariance Matrix

## Definition (Sample Covariance)

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

## Definition (Sample Covariance Matrix)

For a dataset with  $p$  variables, the sample covariance matrix  $S$  is a  $p \times p$  symmetric matrix where the diagonal elements are variances and off-diagonals are covariances:

$$S = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$$

# Example: Covariance and Covariance Matrix

## Example: Calculating Covariance

**Data:**  $X = (1, 2, 3)$ ,  $Y = (2, 4, 6)$

**Means:**  $\bar{x} = 2$ ,  $\bar{y} = 4$

**Covariance ( $s_{xy}$ ):**

$$s_{xy} = \frac{(1-2)(2-4) + (2-2)(4-4) + (3-2)(6-4)}{3-1} = \frac{(-1)(-2) + 0 + (1)(2)}{2} = \frac{4}{2} = 2$$

## Example: Constructing the Matrix

With  $s_x^2 = 1$ ,  $s_y^2 = 4$ , and  $s_{xy} = 2$ , the  $2 \times 2$  covariance matrix is:

$$S = \begin{pmatrix} s_x^2 & s_{xy} \\ s_{xy} & s_y^2 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$$

# Frequentist Statistics: Sampling

## Frequentist Paradigm

Frequentist statistics views probability as the long-run frequency of events. True population parameters (e.g., mean  $\mu$ ) are considered **fixed but unknown**, and sample data are treated as random outcomes from repeated experiments.

## Sampling

Drawing a subset of data from the population to estimate parameters. The estimators themselves are random variables with their own sampling distributions.

## Example: Sampling Distribution

If you repeatedly draw samples of size  $n = 30$  from a population and compute the sample mean  $\bar{x}$  each time, the distribution of these  $\bar{x}$  values forms the **sampling distribution of the mean**.

# Mean Square Error (MSE) & Consistency

## Definition (Mean Square Error (MSE))

MSE measures the average squared difference between an estimator  $\hat{\theta}$  and the true parameter  $\theta$ .

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2] = \text{Variance}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$$

## Definition (Consistency)

An estimator  $\hat{\theta}$  is **consistent** if, as the sample size  $n$  approaches infinity,  $\hat{\theta}$  converges in probability to the true parameter  $\theta$ .

## Example: MSE and Consistency

The sample mean  $\bar{x}$  is an unbiased estimator of  $\mu$ , so its Bias is 0. Its MSE is just its variance,  $\frac{\sigma^2}{n}$ . As  $n \rightarrow \infty$ ,  $\text{MSE} \rightarrow 0$ , demonstrating that  $\bar{x}$  is a **consistent** estimator of  $\mu$  (Law of Large Numbers).

# Confidence Intervals

## Definition (Confidence Interval)

A Confidence Interval (CI) gives an estimated range of values which is likely to include an unknown population parameter. A  $(1 - \alpha)$  confidence level means that if we repeat the sampling many times,  $100(1 - \alpha)\%$  of the constructed intervals will contain the true parameter.

## Example: 95% Confidence Interval for the Mean

Suppose a sample of  $n = 100$  has a mean  $\bar{x} = 50$  and standard deviation  $s = 10$ . The 95% CI is calculated as:

$$CI = \bar{x} \pm 1.96 \left( \frac{s}{\sqrt{n}} \right) = 50 \pm 1.96 \left( \frac{10}{10} \right) = 50 \pm 1.96 = [48.04, 51.96]$$

We are 95% confident that the true population mean lies between 48.04 and 51.96.

# Parametric vs Non-parametric Model Estimation

## Parametric Estimation

Assumes data follows a specific underlying probability distribution (e.g., Normal, Poisson) governed by a fixed set of parameters  $(\mu, \sigma)$ .

## Example

Assuming student heights follow a Normal distribution and using the sample to estimate  $\mu$  and  $\sigma$ . The model is rigid but highly efficient if the assumption holds.

## Non-parametric Estimation

Does not assume a specific parametric family of probability distributions. The model structure is determined from data (infinite-dimensional).

## Example

Using a **Histogram** or **Kernel Density Estimation (KDE)** to visually estimate the probability density function of heights without assuming they are normally distributed.

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